

OAKEY, Australia

WMO: 945520

Lat: 27.4033S

Lon: 151.7414E

Elev: 407

StdP: 96.53

Time Zone: 10.00 (E10)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -2.0	(c) -0.3	(d) -8.0	(e) 2.0	(f) 12.8	(g) -5.8	(h) 2.4	(i) 10.5	(j) 11.3	(k) 15.6	(l) 10.4	(m) 15.5	(n) 2.1	(o) 90	(p) 0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 11.9	(c) 35.3	(d) 19.8	(e) 33.6	(f) 19.6	(g) 32.0	(h) 19.4	(i) 23.0	(j) 28.9	(k) 22.3	(l) 28.1	(m) 21.7	(n) 27.5	(o) 5.5	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	16.9	24.7	20.7	16.2	24.4	20.1	15.5	24.0	70.2	28.8	67.6	28.3	65.4	27.5	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.5	8.7	-4.7	38.3	1.2	1.8	-5.6	39.6	-6.2	40.7	-6.9	41.7	-7.7	43.0
			-5.4	24.2	1.3	0.7	-6.4	24.8	-7.1	25.2	-7.9	25.6	-8.8	26.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	24.4	23.9	22.2	18.7	14.5	11.7	10.9	12.2	16.1	19.4	21.8	23.5
	DBStd	5.55	2.29	2.47	2.19	2.37	2.89	2.94	2.85	2.98	3.17	2.98	2.88	2.79
	HDD10.0	53	0	0	0	0	3	16	23	11	1	0	0	0
	HDD18.3	876	0	0	1	23	122	198	230	192	81	23	4	1
	CDD10.0	3061	447	388	379	260	142	68	52	79	183	291	353	418
	CDD18.3	842	189	155	122	33	3	0	0	2	14	56	108	160
	CDH23.3	7919	1731	1310	960	319	48	3	3	43	291	675	1041	1496
	CDH26.7	2996	741	530	307	48	4	1	0	8	79	232	409	637
Wind	WSAvg	4.4	5.1	5.1	4.7	4.1	3.7	3.7	3.7	4.0	4.3	4.7	5.0	4.9
Precipitation	PrecAvg	638	76	85	47	36	44	27	32	26	31	59	79	98
	PrecMax	901	214	262	109	229	190	136	136	63	105	196	206	188
	PrecMin	335	2	0	0	0	2	0	0	0	0	5	1	13
	PrecStd	160	53	65	33	46	47	28	28	20	27	43	54	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	37.1	34.9	30.0	26.8	23.0	23.5	27.0	31.7	34.5	35.4	37.1
		MCWB	21.0	20.9	19.7	17.7	15.7	14.3	13.7	14.2	16.1	17.0	18.6	19.6
	2%	DB	34.7	34.1	31.7	28.0	24.7	21.4	21.5	24.1	28.9	31.3	33.2	34.5
		MCWB	20.4	21.1	19.8	17.1	15.1	14.0	13.1	12.9	15.1	16.4	18.4	19.4
	5%	DB	32.6	31.9	29.8	26.6	23.1	20.2	20.0	22.1	26.7	29.2	31.0	32.2
		MCWB	20.4	20.7	19.5	16.9	14.6	13.3	12.5	12.3	14.7	16.5	18.0	19.5
	10%	DB	30.7	29.9	28.2	25.1	21.8	18.9	18.5	20.4	24.5	27.1	29.0	30.2
		MCWB	20.3	20.2	19.1	16.7	14.0	12.8	11.7	11.7	14.1	15.9	17.9	19.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	23.8	22.8	21.0	18.5	17.1	17.2	16.6	18.6	20.7	22.0	23.1
		MCDB	30.2	30.2	27.8	25.3	22.3	20.2	20.7	22.8	25.1	26.5	29.2	30.0
	2%	WB	22.9	23.0	21.8	19.4	17.2	15.8	15.3	15.3	17.4	19.4	21.0	22.2
		MCDB	28.9	28.7	27.1	24.3	21.4	19.0	18.5	20.4	23.7	25.2	27.8	28.5
	5%	WB	22.2	22.2	21.1	18.5	16.3	14.8	14.0	14.2	16.4	18.3	20.1	21.5
		MCDB	28.2	27.6	26.1	23.5	20.5	18.2	17.4	19.0	22.8	24.4	26.5	27.6
	10%	WB	21.6	21.5	20.4	17.7	15.5	14.0	13.0	13.1	15.5	17.5	19.3	20.9
		MCDB	27.6	26.9	25.4	22.7	19.7	17.3	16.9	18.2	21.6	23.7	25.6	26.9
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.4	11.8	13.2	14.5	14.0	15.2	16.5	16.0	14.3	13.0	12.3
		MCDBR	15.3	14.7	14.6	15.2	16.2	15.5	16.7	18.7	19.0	17.9	16.4	15.7
		MCWBR	4.5	4.2	4.7	6.1	8.0	8.6	9.3	9.4	7.7	6.5	5.2	4.8
	5% WB	MCDBR	11.8	11.4	11.2	11.7	11.7	11.5	12.0	13.9	14.5	13.3	12.6	12.2
		MCWBR	4.2	3.9	4.3	5.2	6.0	6.8	7.4	7.6	6.6	5.5	4.7	4.4
Clear-Sky Solar Irradiance	taub	0.364	0.356	0.334	0.317	0.296	0.290	0.283	0.295	0.330	0.353	0.363	0.372	
	taud	2.529	2.559	2.623	2.639	2.647	2.670	2.673	2.611	2.501	2.471	2.483	2.485	
	Ebn at Noon	978	969	962	933	912	897	919	944	950	961	973	972	
	Edh at Noon	112	106	94	84	76	70	73	85	104	114	116	117	
All-Sky Solar Radiation	RadAvg	6.85	6.27	5.46	4.72	3.88	3.21	3.67	4.62	5.66	6.33	6.86	6.88	
	RadStd	0.54	0.57	0.34	0.22	0.20	0.25	0.28	0.34	0.44	0.60	0.64	0.68	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-66	+179	+109

Nomenclature: See separate page